

Experiment 2

Aim : Write a program to implement inter process communication.

Theory :

- Interprocess communication in distributed system is a process of exchanging data between two or more independent processes in a distributed environment called Inter-process communication.
- Interprocess communication on the internet provides both Datagram and stream communication.
- Characteristics :

1) Synchronous system calls

⇒ In this system calls both sender and receiver use blocking system calls to transmit the data.

2) Asynchronous system calls

⇒ In this system calls, both sender and receiver ~~waits until the message are~~ use non-blocking system call to transmit the data.

3) Reliability

⇒ It is defined as validity and integrity.

- Types:

1) Message Passing

⇒ Message passing involves process communicating by sending and receiving messages. Messages can be structured data packets containing information or commands.

2) RPC

⇒ RPC allows one process to invoke a procedure in another process, typically located on a different machine over a network.

- Advantages:

1) Enables processes to communicate with each other and share resources.

2) Can be used to implement various synchronization.

- Disadvantages:

1) Increases system complexity.

2) Can introduce security vulnerabilities.

